

QLINIC DRILL SERIES

(Precision Drills for Perfect Score)

UNIBEN POST-UTME PAST QUESTIONS

COMPILED BY Etionsa Aiwanseido (A.JDaniel)



What to know about this Pack before studying: This Question Bank is a collection of Original (2021 – 2024) Questions which were gotten from interviewing students who wrote the exam during these years. Some of the options were re-modified professionally, as many of them could not remember all four options. However, the question and the correct answer are authentic. During the production of this booklet we ensure to avoid temptation of introducing any new question as that will alter our main vision of being authentic and reliable. However, it is important to let you know that studying the booklet alone is not an ideal strategy as exam patterns can be changed anytime.

(Prescription)How to maximise your chances properly: it is highly recommended you study your O-level textbooks, Notes, along side with QLINIC DRILL SERIES (QDS PACK).

Structure of the exam: As at 2019 – 2022, the exam was a CBT based exam with 50 question's in total for 1hour, it was usually 20 English, 10 Physics, Chemistry, Biology or Mathematics each, depending on your UTME Combination. But from 2023-2024 the exam Structure changed, 2023, the exam was 30 Questions in 30mins, where one question was about 3.33Mark, but in 2024 it was 30mins for 40 question where you have 12 English, 12 Biology, 8 Physics, 8 Chemistry. This year it is more likely to change, so prepare by working with limited time frame to be equipped.

Nature of Questions: The Exam is actually O-Level Based, it will be ridiculous to be asked Advance Level questions when you are not in that spot yet. However they are times where you can see some tough questions maybe one or two, that may not be easily found in some of your basic textbooks, still this is not a reason to drop your basic textbooks and start reading 100lv text, you are cheating your chances.

How to participate for this entrance exam: You should score above 200 In UTME, and Make Uniben your first choice. After which you can purchase and register online for the Post-UTME When the form is out. **You can contact Us for your swift registration, at a cheaper rate, while relaxing at home.** You do not need to come down to Uniben for registration, you are only to be here during the exam.



Biology Questions

(1) Which of the following organelles is known as the kitchen house of the cell?

- A. Mitochondrion
- B. Endoplasmic reticulum
- C. Chloroplast
- D. Golgi bodies

[2021]

(2) Which of the following vertebrae prevents sagging?

- A. Lumbar
- B. Thoracic
- C. Claudel
- D. Caudal

[2021]

(3) Which of the following cell organelles carries out cell specialization?

- A. Nucleus
- B. Cytoplasm
- C. Microtubules
- D. Mitochondrion

[2022]

(4) Which part of the tooth is sensitive to cold and hot?

- A. Cement
- B. Pulp cavity
- C. Dentine
- D. Periodontal membrane

[2020]

(5) Monera do not have true — ?

- A. Nuclei
- B. Cytoplasm
- C. Membrane

D. Root

[2021]

(6) Which of the following is not found under the phylum Coelenterata?

- A. Hydra
- B. Jellyfish
- C. Sea anemone
- D. Sea urchin

[2022]

(7) The bicuspid valve is located between

- A. Right auricle and right ventricle
- B. Left ventricle and right auricle
- C. Right ventricle and right auricle
- D. Left ventricle and left auricle

[2024]

(8) Which of the following is a trace element?

- A. Fe
- B. Na
- C. Cl
- D. Mn

[2023]

(9) What are the organs found in the stomach?

- A. Kidney, spleen, gall bladder
- B. Liver, kidney, gall bladder
- C. Pancreas, gall bladder, spleen
- D. Liver, kidney, spleen

[2024]

(10) If a tree plant grows up without being attached to the ground, it is found in which biome?

- A. Savannah



- B. Rainforest
- C. Grassland
- D. Tropical Rainforest

[2024]

(11) In a plant, if the epicotyl grows before the hypocotyl, it is regarded as what germination?

- A. Epigeal
- B. Hypogeal
- C. Viviparous
- D. Ovoviviparous

[2024]

(12) When the eyeball is too short, what type of eye defect occurs?

- A. Hypermetropia
- B. Myopia
- C. Presbyopia
- D. Astigmatism

[2024]

(13) What is used to correct presbyopia?

- A. Converging lens
- B. Diverging lens
- C. Biconcave lens
- D. Cylindrical lens

[2024]

(14) The ability to taste P.T.C is an example of what type of variation?

- A. Continuous variation
- B. Discontinuous variation
- C. Morphological variation
- D. Environmental variation

[2024]

(15) What term is used in the expression of a gene?

- A. Genotype
- B. Phenotype
- C. Archetype
- D. Genome

[2024]

(16) Which of the following is not an evidence that supports evolution?

- A. Pathology
- B. Serology
- C. Paleontology
- D. Vestigial structures

[2024]

(17) Plants found in water often lack which structural adaptation?

- A. Well-developed roots
- B. Stomata on the lower epidermis
- C. Cuticle
- D. Vascular tissues

[2024]

(18) What helps most in refraction of light rays in the eye?

- A. Lens and cornea
- B. Lens and vitreous humor
- C. Cornea and vitreous humor
- D. Cornea and aqueous humor

[2024]

(19) What closes in the larynx during swallowing?

- A. Epiglottis
- B. Glottis
- C. Trachea
- D. Bronchi

[2024]

(20) Which of the following structures



is not a part of a fern plant?

- A. Rhizoid
- B. Rhizome
- C. Sporangium
- D. Fronds

[2024]

(21) The division of the nucleus is called?

- A. Cell division
- B. Karyokinesis
- C. Cytokinesis
- D. Mitosis

[2024]

(22) Which of the following plants is commonly found in the savannah?

- A. Baobab
- B. Acacia
- C. Date palm
- D. Both A and B

[2024]

(23) What is a key difference between bony fish and cartilaginous fish?

- A. Streamlined body
- B. Gills
- C. Bladder
- D. Lateral line

[2024]

(24) Wooden and stunted growth occurs in which biome?

- A. Savannah
- B. Shrub
- C. Mangrove
- D. Tundra

[2024]

(25) Which of the following is an example of drupe and berry?

- A. Mango and Tomatoes
- B. Orange and Grapes
- C. Apple and Banana
- D. Peach and Watermelon

[2024]

(26) Asexual stage of Plasmodium occurs where?

- A. Mosquito
- B. Man
- C. Water
- D. Soil

[2024]

(27) The female and male parts of a flower are called?

- A. Stamen and Pistil
- B. Petals and Sepals
- C. Roots and Stems
- D. Leaves and Branches

[2024]

(28) The possession of pneumatophores is a characteristic feature of which biome?

- A. Tropical rainforest
- B. Grassland
- C. Freshwater swamp
- D. Mangrove swamp

[2024]

(29) What pigment is primarily responsible for the red, purple, and blue coloration in plants?

- A. Chlorophyll
- B. Carotene



C. Anthocyanin

D. Xanthophyll

[2024]

(30) When an athlete has finished running, which of the following processes helps in cooling the body?

A. Radiation

B. Conduction

C. Convection

D. Perspiration

[2024]

(31) Which of the following diseases is not associated with the kidney?

A. Nephritis

B. Kidney stones

C. Hepatitis

D. Uremia

[2024]

(32) Which of the following blood groups is known as the universal donor?

A. O+

B. AB-

C. A+

D. O-

[2024]

(33) The process of growing microorganisms under controlled laboratory conditions is called:

A. Inoculation

B. Incubation

C. Fermentation

D. Culturing

[2024]

(34) The profile diagram of a

geographical landscape is largely determined by:

A. Vegetation cover

B. Human activities

C. Climatic factors

D. Topography

[2024]

(35) The process by which seeds absorb water to initiate germination is called:

A. Diffusion

B. Imbibition

C. Capillarity

D. Transpiration

[2024]

(36) Hydra belongs to which of the following phyla?

A. Platyhelminthes

B. Annelida

C. Coelenterata

D. Mollusca

[2024]

(37) Meiosis is best described as a process of:

A. $2n \rightarrow n$

B. $n \rightarrow n$

C. $n \rightarrow 2n$

D. $2n \rightarrow 2n$

[2024]

(38) The nerve in the teeth is located in the:

A. Dentine

B. Enamel

C. Pulp cavity

D. Cementum



[2024]

(39) Feeding in Paramecium is carried out through the:

- A. Cilia
- B. Contractile vacuole
- C. Oral groove
- D. Flagella

[2024]

(40) Planula is the larval stage of:

- A. Earthworm
- B. Spirogyra
- C. Hydra
- D. Amoeba

[2024]

(41) Which of the following plants does not undergo secondary growth?

- A. Cocoa
- B. Palm tree
- C. Mango
- D. Orange tree

[2024]

(42) The membrane that encloses and protects the heart is called the:

- A. Endocardium
- B. Myocardium
- C. Pericardium
- D. Epicardium

[2024]

(43) Which of the following states can be found in the Guinea savanna zone of Nigeria?

- A. Lagos
- B. Borno
- C. Kwara

D. Rivers

[2024]

(44) The alimentary canal of man can be infested with:

- A. Plasmodium falciparum
- B. Ascaris lumbricoides
- C. Trypanosoma brucei
- D. Entamoeba gingivalis

[2024]

(45) Evolution between flowering plants and their pollinators is an example of:

- A. Convergent evolution
- B. Divergent evolution
- C. Parallel evolution
- D. Co-evolution

[2024]

(46) The feeding structure found in a snail is called the:

- A. Mandible
- B. Spiracle
- C. Radula
- D. Proboscis

[2024]

(47) During anaerobic respiration in plants, the products formed are:

- A. Ethanol and water
- B. Ethanol and carbon dioxide
- C. Lactic acid and water
- D. Oxygen and carbon dioxide

[2024]

(48) Which of the cell organelles carries out cell specialisation in living organisms?



- A. Nucleus
- B. Mitochondrion
- C. Cytoplasm
- D. Mesosomes

(49) The kind of vision where the eye is able to see two objects at the same time is known as

- A. Visual perception
- B. Stereoscopic vision
- C. Monocular vision
- D. Binocular vision

(50) Which of the following is associated with dead cells

- A. wood fibre
- B. Parenchyma
- C. Collenchyma
- D. Sclerenchyma

51. The red coloration of the soil is due to the presence of

- A. Magnesium oxide
- B. Copper sulphide
- C. Ferric oxide (Fe_2O_3)
- D. Aluminium hydroxide

52. Guard cells have all the following except

- A. Mitochondria
- B. Nucleus
- C. Chloroplast
- D. Centrioles

53. The immune response of the skin is carried out by

- A. Schwann cells

- B. Mast cells
- C. Keratinocytes
- D. Langerhans cells

54. Which of the following pathogens causes cassava mosaic disease?

- A. Fungi
- B. Nematodes
- C. Viruses
- D. Viroids

55. In Mucor or Rhizopus, carbohydrate is absorbed in the form of

- A. Starch
- B. Sucrose
- C. Glucose
- D. Glycogen

56. Which of the following does not undergo secondary growth?

- A. Mango
- B. Palm tree
- C. Mahogany
- D. Cocoa

57. Which of the following is not found in the prothallus of a fern?

- A. Capsule
- B. Androecium
- C. Archegonium
- D. Rhizoid

58. The shedding of leaves in plants is referred to as



- A. Senescence
- B. Defoliation
- C. Abscission
- D. Translocation

59. Which organism is territorial?

- A. *Tilapia nilotica*
- B. *Agama agama*
- C. *Paramecium caudatum*
- D. *Trypanosoma brucei*

60. The structure that carries impulses away from the nerve cell is

- A. Dendrite
- B. Axon
- C. Synapse
- D. Soma

61. Cassava mosaic disease is transmitted by

- A. *Heteroligus meles*
- B. *Maruca testulalis*
- C. *Sesamia spp.*
- D. *Bemisia nigeriensis*

62. Onions store food in their

- A. Corm
- B. Stem
- C. Leaves
- D. Root

63. Which of the following is NOT true about *Mucor* or *Rhizopus*?

- A. Grows on moist dead organic matter

B. Is a plant

C. Has cellulose cell wall

D. Reproduces asexually by spores

64. The evidence that supports the advancement of ferns over mosses is based on

- A. Comparative anatomy
- B. Molecular records
- C. Biochemical similarities
- D. Physiological records

65. Which of the following hormones regulates the opening and closing of stomata?

- A. Cytokinin
- B. Absciscic acid
- C. Auxin
- D. Gibberellin

66. The seed contains all of the following except

- A. Endosperm
- B. Embryo sac
- C. Exocarp
- D. Testa

67. Tse-tse fly causes sleeping sickness, also known as

- A. Filariasis
- B. Trypanosomiasis
- C. Malaria
- D. Onchocerciasis

Chemistry Questions



1. What gas causes global warming of the earth?
 - A. CO_2
 - B. SO_2
 - C. CFCs
 - D. Ozone
2. What is the trivial name for butadenoic acid?
 - A. Formic acid
 - B. Maleic acid
 - C. Butyric acid
 - D. Succinic acid
3. What is Molanic acid?
 - A. Ethanoic acid
 - B. Propanoic acid
 - C. Ethanedioic acid
 - D. Prop-2-enoic acid
4. Petroleum coke is:
 - A. Peat
 - B. Bitumen
 - C. Anthracite
 - D. Residual carbon
5. Which of the following is NOT produced during the destructive distillation of coal?
 - A. Coke
 - B. Ammoniacal liquor
 - C. Benzene
 - D. Coal fibre
6. Which of the following gases cannot be collected over water?
 - A. HCl
 - B. CO
 - C. NH_3
 - D. SO_2
7. What are the components of tool steel?
 - A. Iron, Carbon, Chromium
 - B. Iron, Nickel, Lead
 - C. Iron, Carbon, Tungsten
 - D. Iron, Aluminium, Zinc
8. What is the catalyst used in the Contact Process?
 - A. MnO_2
 - B. Fe_2O_3
 - C. V_2O_5
 - D. CuO
9. Which of the following compounds does NOT contain a double bond?
 - A. Propene
 - B. Ethene
 - C. Propane
 - D. Ethanol
10. The scattering of light in a colloidal solution is called:
 - A. Brownian effect
 - B. Compton effect
 - C. Tyndall effect
 - D. Doppler effect
11. In $\text{C}=\text{O}$, the type of bond present is:
 - A. Ionic
 - B. Covalent
 - C. Dative
 - D. Metallic
12. What is the correct order of increasing oxidation strength among halogens?
 - A. $\text{F} > \text{Cl} > \text{Br} > \text{I}$



B. $I > Br > Cl > F$

C. $Cl > F > I > Br$

D. $Br > I > Cl > F$

13. Which of the following is the formula for water glass?

A. Na_2SiO_3

B. K_2SiO_3

C. $Na_2Si_2O_5$

D. Na_2SO_4

14. When calcium is heated, it emits light known as:

A. Solar light

B. Tube light

C. Arc light

D. Lime light

15. Which of the following is used to dissolve bauxite in its extraction?

A. HCl

B. HNO_3

C. NaOH

D. H_2SO_4

16. The IUPAC name for capric acid is:

A. Octanoic acid

B. Decanoic acid

C. Dodecanoic acid

D. Nonanoic acid

17. What is the molecular formula of an alkane with a vapor density of 15?

A. C_2H_6

B. CH_4

C. C_3H_8

D. C_4H_{10}

18. Which allotrope of carbon is used in making plastic composites?

A. Diamond

B. Carbon fibre

C. Graphite

D. Fullerene

19. During the production of ammonia, the catalyst used is:

A. Iron catalyst

B. Finely divided nickel

C. Granulated iron

D. Platinum

20. The catalyst used in the oxidation of SO_2 to SO_3 is:

A. Fe_2O_3

B. NiO

C. V_2O_5

D. MnO_2

21. Which of the following compounds decomposes upon heating?

A. K_2SO_4

B. Na_2CO_3

C. K_2CO_3

D. $NaHCO_3$

22. Which of the following can be used to sterilise hospital equipment?

A. Ultraviolet rays

B. Infrared rays

C. Gamma rays

D. X-rays

23. Which of these gases can be collected over water?

A. O_2

B. CO_2

C. NH_3

D. HCl



24. Formic acid can be obtained from:

- A. Methane
- B. Red ants
- C. Banana peel
- D. Sulfur dioxide

25. What is the correct order of increasing lattice energy for sodium halides?

- A. $\text{NaI} < \text{NaCl} < \text{NaBr} < \text{NaF}$
- B. $\text{NaI} < \text{NaBr} < \text{NaCl} < \text{NaF}$
- C. $\text{NaF} < \text{NaCl} < \text{NaBr} < \text{NaI}$
- D. $\text{NaBr} < \text{NaI} < \text{NaCl} < \text{NaF}$

26. What is the formula for salt cake soda?

- A. Na_2SO_4
- B. Na_2CO_3
- C. NaHCO_3
- D. NaCl

27. Which of the following is NOT true about halogens?

- A. Atomic radius decreases down the group
- B. They exist in different physical states at room temperature
- C. Electronegativity decreases down the group
- D. Reactivity increases steadily down the group

28. What is the molecular formula of a compound with empirical formula $(\text{CH}_6\text{O})_n$ and molar mass of 64 g/mol?

- A. $\text{C}_2\text{H}_{12}\text{O}_2$
- B. CH_6O
- C. $\text{C}_3\text{H}_{18}\text{O}_3$

D. $\text{C}_4\text{H}_{24}\text{O}_4$

29. Photographic flashlight is made up of two gases. Which are they?

- A. Krypton and Helium
- B. Helium and Oxygen
- C. Krypton and Xenon
- D. Helium and Neon

30. A gas is said to behave like an ideal gas under

- A. Low temperature and high pressure
- B. High pressure and low density
- C. High temperature and low pressure
- D. Low molar mass and high compressibility

31. The solubility of AgCl is $1.35 \times 10^{-5} \text{ mol/dm}^3$ in pure water at 25°C .

Calculate the solubility product (K_{sp}) of the salt.

- A. $1.82 \times 10^{-10} \text{ mol}^2/\text{dm}^6$
- B. $1.82 \times 10^{-9} \text{ mol}^2/\text{dm}^6$
- C. $1.35 \times 10^{-10} \text{ mol}^2/\text{dm}^6$
- D. $1.35 \times 10^{-9} \text{ mol}^2/\text{dm}^6$

32. Which noble gas is radioactive?

- A. Neon
- B. Xenon
- C. Argon
- D. Radon

33. The solubility product of a sparingly soluble salt MX_2 is $6.0 \times 10^{-9} \text{ mol}^3/\text{dm}^9$ at 25°C . What is the solubility of the salt at 25°C ?

- A. $3 \times 10^{-3} \text{ mol/dm}^3$
- B. $3 \times 10^{-5} \text{ mol/dm}^3$
- C. $6 \times 10^{-3} \text{ mol/dm}^3$



D. $6 \times 10^{-2} \text{ mol/dm}^3$

34. The two pairs of gases used by sea divers are

- A. Helium and Oxygen
- B. Nitrogen and Oxygen
- C. Neon and Helium
- D. Krypton and Oxygen

35. Which noble gas is used in aerodrome beacons?

- A. Neon
- B. Argon
- C. Xenon
- D. Helium

36. What is the pH value of blood?

- A. 6.9
- B. 7.4
- C. 4.5
- D. 7.0

37. Abundant noble gas in the atmosphere is

- A. Neon
- B. Krypton
- C. Argon
- D. Helium

38. How many percentage of phosphorus is in the human body?

- A. 1.0%
- B. 0.5%
- C. 0.32%
- D. 0.41%

39. Which of the following does not form amphoteric oxides?

- A. Fe
- B. Al

C. Pb

D. Zn

40. Function of radon is

- A. Treatment of cancer
- B. Manufacture of light bulbs
- C. Welding gases
- D. Food preservation

41. What is the confirmatory test for the presence of water?

- A. Blue cobalt(II) chloride turning white
- B. Blue copper(II) sulfate turning green
- C. Anhydrous zinc chloride turning grey
- D. Blue copper(II) chloride turning pink

42. Sodium hydroxide is used in qualitative analysis to test for which of the following ions?

- A. Ca^{2+}
- B. NH_4^+
- C. Zn^{2+}
- D. Cu^{2+}

43. Which of the following does not involve the destructive distillation of coal?

- A. Coal tar
- B. Ammoniacal liquor
- C. Benzene
- D. Coal fiber

44. The general molecular formula of the alkane homologous series is

- A. C_nH_{2n}
- B. $\text{C}_n\text{H}_{2n-2}$



C. C_nH_{2n+2}

D. $C_nH_{2n}O_2$

45. Which of the following exhibits both vibrational and translational motion?

A. Stone

B. Water

C. Ammonia gas

D. Alcohol

46. The main components of formalin include

A. 40% methanol and 60% water

B. 37% formaldehyde and 63% water

C. 20% ethanol and 80% methanol

D. 25% acetone and 75% water

47. Which organic acid is commonly used to lower blood pressure?

A. Formic acid

B. Acetic acid

C. Tartaric acid

D. Ethanoic acid

48. Vapour pressure (SVP) depends primarily on

A. Vapour content

B. Pressure variation

C. Temperature

D. Volume

49. The electrolyte used in electroplating an article with silver is

A. Silver nitrate solution

B. Silver cyanide solution

C. Sodium argentocyanide solution

D. Nickel sulfate solution

50. Aluminium powder is used in thermite welding because

A. It is corrosion resistant

B. It is a good conductor of heat

C. It is a strong reducing agent

D. It absorbs heat during reaction

51. The metals zinc and tin are found in which alloy?

A. Brass

B. Bronze

C. Duralumin

D. Solder

52. Which of the following metals is not present in stainless steel?

A. Chromium

B. Carbon

C. Nickel

D. Aluminium

53. This is NOT an alloy of copper:

A. Brass

B. Bronze

C. Solder

D. Duralumin

54. Which process is used in the industrial production of ammonia (NH_3)?



- A. Ostwald Process
- B. Contact Process
- C. Solvay Process
- D. Haber Process

- A. Cathode
- B. Anode
- C. Salt bridge
- D. Electrolyte

55. Another name for wood spirit is

- A. Ethanol
- B. Methanol
- C. Propanol
- D. Butanol

60. Another name for malonic acid is

- A. Ethanoic acid
- B. Butanoic acid
- C. Ethanedioic acid
- D. Propanedioic acid

56. Which pair of metals are the major components of bronze?

- A. Copper and zinc
- B. Copper and tin
- C. Copper and nickel
- D. Copper and lead

Physics Questions

(1) During floatation of a body, the forces involved are

- A. Upthrust and viscous force
- B. Upthrust and weight force
- C. Upthrust and tension
- D. Upthrust, viscous and tension force

[22/23]

57. Which of the following is a saturated hydrocarbon compound?

- A. C_3H_5
- B. C_3H_6
- C. C_2H_5Cl
- D. C_3H_8

(2) Anomalous expansion of water was discovered by

- A. Helen
- B. Galileo
- C. Hope
- D. Newton

[2023]

58. The property used to separate groundnut oil from water is

- A. Boiling point
- B. Solubility
- C. Density
- D. Rate of diffusion

(3) The process whereby an oscillating system gradually loses energy and comes to rest is called

- A. Forced vibration
- B. Damping
- C. Resonance
- D. Energy oscillation

[2023]

59. In electrolytic refining of metals, the impure metal is made the

(4) Which of the following does a



spring balance work on?

- A. Principle of moments
- B. Newton's first law
- C. Hooke's law
- D. Pascal's law

[2023]

(5) A fuel-driven motor, which obtains high voltage from a low one is

- A. A.C. dynamo
- B. D.C. generator
- C. Transformer
- D. Battery inverter

[2024]

(6) If $V = 5 + 3t^2$, calculate the acceleration when $t = 2s$

- A. 6 m/s^2
- B. 9 m/s^2
- C. 12 m/s^2
- D. 15 m/s^2

[2024]

(7) The density of a liquid is 900 kg/m^3 and its weight is 60 g. What is the weight of the same volume of liquid when the density becomes 800 kg/m^3 ?

- A. 66.7 g
- B. 53.3 g
- C. 48.0 g
- D. 60.0 g

[2024]

(8) What is the dimension of power?

- A. ML^2T^{-3}
- B. ML^2T^{-2}
- C. MLT^{-2}
- D. MLT^{-3}

[2023]

(9) The principle by which a transformer works is

- A. Electromagnetic induction
- B. Mutual inductance
- C. Self-induction
- D. Ohm's law

[2023]

(10) Which of the following instruments measures both AC and DC current?

- A. Galvanometer
- B. Moving iron meter
- C. Ammeter
- D. Voltmeter

[2023]

(11) Air is a good ___ of heat

- A. Insulator
- B. Conductor
- C. Radiator
- D. Moderator

[2022]

(12) Which of the following instruments is most appropriate for measuring the thickness of a sheet of paper?

- A. Spherometer
- B. Meter rule
- C. Micrometer screw gauge
- D. Vernier caliper

[2024]

(13) If silicon is doped with phosphorus, what type of semiconductor material is formed?



A. Zener material

B. P-N junction

C. N-type

D. P-type

[2024]

14. The resultant of three vectors is maximum when their angle of inclination is

A. 180°

B. 90°

C. 45°

D. 0°

15. The Resultant of two vectors is minimum when they are

A. Parallel

B. Anti-parallel

C. Perpendicular

D. Coplaner

16. The following are found in an electric bell except __?

A. Solenoid

B. Contact breaker

C. Permanent magnet

D. Electromagnets

17. A charge of 10 C moves through a potential difference of 12 V. What is the work done?

A. 1.2 J

B. 22 J

C. 120 J

D. 0.83 J

18. Which of the following best describes relative velocity?

A. Velocity measured from the ground

B. Velocity of one object with respect to another

C. Sum of the individual velocities

D. Product of speed and time

19. A lever has an effort arm of 2.0 m and a load arm of 0.5 m. What is the mechanical advantage of the lever?

A. 0.25

B. 2.5

C. 4.0

D. 0.50

20. A radioactive isotope decays from 80 g to 10 g in 3 hours. What is the half-life?

A. 30 mins

B. 1 hour

C. 45 mins

D. 90 mins

21. The unit of specific heat capacity is:

A. J/kg

B. J/kg $^\circ$ C

C. J/kg/m

D. J/ $^\circ$ C

22. Which electromagnetic wave has the highest frequency?

A. Radio waves

B. Gamma rays

C. Infrared



D. Microwaves

23. Calculate the pressure at the bottom of a tank 10 m deep filled with water (density = 1000 kg/m^3 , $g = 10 \text{ m/s}^2$).

- A. 10,000 Pa
- B. 100,000 Pa
- C. 1,000 Pa
- D. 100 Pa

24. A 2 kg block of copper is heated from 20°C to 80°C . (Specific heat capacity of copper = $400 \text{ J/kg}^\circ\text{C}$). How much heat is absorbed?

- A. 48,000 J
- B. 24,000 J
- C. 32,000 J
- D. 12,000 J

25. A sample has a half-life of 5 days. How much of a 64 g sample remains after 15 days?

- A. 8 g
- B. 32 g
- C. 4 g
- D. 16 g

26. Which of these does not affect the rate of heat transfer?

- A. Temperature difference
- B. Surface area
- C. Colour of the object
- D. Volume of the object

27. Which statement is true of pressure in a fluid at rest?

- A. It increases with height
- B. It is independent of depth
- C. It is equal in all directions at the same depth
- D. It depends only on volume

28. The acceleration due to gravity on the surface of the Earth is approximately:

- A. 10 m/s^2
- B. 9.8 m/s
- C. 8.0 m/s^2
- D. 6.67 m/s^2

29. What is the density of a substance with a mass of 500 g and volume of 200 cm^3 ?

- A. 2.5 g/cm^3
- B. 0.25 g/cm^3
- C. 0.40 g/cm^3
- D. 2.0 g/cm^3

30. A machine with a velocity ratio of 5 has an efficiency of 80%. Calculate the mechanical advantage.

- A. 4.0
- B. 6.25
- C. 0.25
- D. 3.2

31. A radio wave travels at a speed of m/s and has a frequency of Hz. What is its wavelength?



- A. 200 m
- B. 100 m
- C. 300 m
- D. 400 m

32. Which of the following phenomena best supports the wave nature of light?

- A. Photoelectric effect
- B. Compton scattering
- C. Interference
- D. Emission spectra

33. A 240 V, 60 W bulb operates for 5 hours. How much energy is consumed?

- A. 1080 kJ
- B. 1200 J
- C. 300 Wh
- D. 720 kJ

34. What is the final temperature when 200 g of water at 30°C is mixed with 300 g of water at 70°C?

- A. 50°C
- B. 60°C
- C. 40°C
- D. 45°C

35. Which of the following is true about gravitational field strength?

- A. It increases with altitude
- B. It is constant everywhere
- C. It decreases with distance from Earth
- D. It is independent of mass

36. In an AC circuit, the current:

- A. Is constant in one direction
- B. Alternates in direction
- C. Has no frequency
- D. Is zero

37. A force of 100 N is applied over an area of 2 m². What is the pressure?

- A. 200 N/m²
- B. 50 N/m²
- C. 20 N/m²
- D. 100 N/m²

38. A boat moves with a velocity of 4 m/s in still water. If the river flows at 3 m/s, what is the resultant velocity of the boat moving downstream?

- A. 7 m/s
- B. 5 m/s
- C. 1 m/s
- D. 12 m/s

39. The photoelectric effect shows that:

- A. Light behaves only as a wave
- B. Light energy is continuous
- C. Light has particle-like properties
- D. Electrons are positively charged

40. A machine lifts a load of 200 N through a vertical height of 5 m using an effort of 100 N over a distance of 15 m. What is the efficiency of the machine?

- A. 66.7%
- B. 75.0%



C. 33.3%

D. 88.9%

41. The unit of inductive reactance is:

A. Henry

B. Ohm

C. Volt

D. Ampere

42. If a photon has energy J , what is its frequency? (Planck's constant = $J \cdot s$)

A. Hz

B. Hz

C. Hz

D. Hz

43. Three resistors of 2Ω , 3Ω , and 5Ω are connected in series. What is their total resistance?

A. 30Ω

B. 10Ω

C. 5.5Ω

D. 1Ω

44. If $336,000 \text{ J}$ of heat is required to vaporize 1.0 kg of water at 100°C , what is the specific latent heat of vaporization of water?

A. $2.1 \times 10^5 \text{ J/kg}$

B. $3.36 \times 10^5 \text{ J/kg}$

C. $4.18 \times 10^3 \text{ J/kg}$

D. $1.00 \times 10^6 \text{ J/kg}$

45. Calculate the power dissipated in a 5Ω resistor carrying 4 A of current.

A. 20 W

B. 100 W

C. 80 W

D. 60 W

46. Which of these quantities is a scalar?

A. Pressure

B. Force

C. Velocity

D. Acceleration

47. A current of 2 A flows through a 3Ω resistor. What is the voltage across it?

A. 6 V

B. 1.5 V

C. 0.67 V

D. 12 V

48. A machine is said to be ideal if:

A. Its efficiency is zero

B. There is always energy loss

C. Efficiency is less than 100%

D. No energy is lost in the process

49. A radioactive substance has a half-life of 2 years . How many years will it take to reduce to $1/16\text{th}$ of its original amount?

A. 2 years

B. 4 years

C. 6 years

D. 8 years

50. Electromagnetic waves are:

A. Longitudinal



- B. Transverse
- C. Circular
- D. Mechanical

English Questions

1. Correct spelling:

- A. Guerilla
- B. Guerrilla
- C. Gorilla
- D. Gerila

2. Correct spelling:

- A. Maintenance
- B. Maintanance
- C. Maintainance
- D. Maintainence

3. To look out for means:

- A. To visit
- B. To chase
- C. To watch carefully
- D. To disregard

4. Synonyms of "Juggernaut":

- A. Silent partner
- B. Unstoppable force
- C. Gentle force
- D. Reluctant power

5. Antonym of "Undeniable":

- A. Disputable
- B. Clear
- C. Obvious
- D. Factual

6. Correct spelling:

- A. Hippopotamus
- B. Hippothamus
- C. Hipopotamus

- D. Hippotamus

7. What is the stressed syllable in Advertisement?

- A. ad-ver-tise-MENT
- B. AD-ver-tise-ment
- C. ad-VER-tise-ment
- D. ad-ver-TISE-ment

8. Correct spelling:

- A. Correspondence
- B. Correspence
- C. Correspondants
- D. Correspondance

9. Synonyms of "Derailed":

- A. Fixed
- B. Completed
- C. Disrupted
- D. Stabilized

10. Antonym of "Determination":

- A. Weakness
- B. Willpower
- C. Persistence
- D. Resolve

11. What is the stressed syllable in Practical?

- A. prac-ti-CAL
- B. PRAC-ti-cal
- C. prac-TI-cal
- D. prack-TIC-al

12. Synonyms of "Pertinent":

- A. Useless
- B. Applicable
- C. Irrelevant
- D. False

13. Correct spelling:



A. Aggresion

B. Agression

C. Agresion

D. Aggression

14. What is the stressed syllable in Democratic?

A. DEM-o-cratic

B. demo-CRA-tic

C. de-MO-cratic

D. demo-crat-IC

15. Synonyms of "Reprimanded":

A. Ignored

B. Celebrated

C. Scolded

D. Praised

16. To be tedious means:

A. To be boring

B. To be exciting

C. To be funny

D. To be violent

17. Antonym of "Stimulate":

A. Depress

B. Motivate

C. Encourage

D. Excite

18. What is the stressed syllable in Accommodation?

A. ac-COM-mo-da-tion

B. AC-com-mo-da-TION

C. accom-mo-da-TION

D. accom-mo-DA-tion

19. Correct spelling:

A. Suprise

B. Surprise

C. Surprize

D. Surperise

20. Which of the following has the same consonant sound as the capitalized part in "baSH"?

A. Rash

B. Bashful

C. Pleasure

D. She

21. Correct spelling:

A. Mischievous

B. Mischievious

C. Mischivous

D. Mishevious

22. Correct spelling:

A. Brassiere

B. Brasier

C. Brasierre

D. Brazier

23. Antonym of "Condemn":

A. Vindictive

B. Absolve

C. Acquit

D. Clear

24. To see red means:

A. To be embarrassed

B. To be angry

C. To be confused

D. To be in love

25. What is the stressed syllable in Omnivorous?

A. om-ni-vor-OUS

B. OM-ni-vor-ous

C. om-NI-vor-ous



D. om-ni-VOR-ous

26. Correct spelling:

A. Corroborate

B. Coroborate

C. Corraborate

D. Coroborrare

27. Antonym of "Impetuous":

A. Thoughtful

B. Rash

C. Impulsive

D. Hasty

28. Synonyms of "Emphasis":

A. Insistence

B. Avoidance

C. Unimportance

D. Neglect

29. Synonyms of "Erratic":

A. Unstable

B. Predictable

C. Regular

D. Consistent

30. What is the stressed syllable in Judiciary?

A. JU-diciary

B. judi-ci-AR-y

C. judi-CI-ary

D. ju-DI-ciary

31. Antonym of "Suspect":

A. Trust

B. Accuse

C. Mistrust

D. Doubt

32. Correct spelling:

A. Authentic

B. Authentic

C. Authintic

D. Authentik

33. What is the stressed syllable in

Omniscient?

A. om-NI-scient

B. OM-ni-scient

C. om-ni-scient

D. om-ni-SCI-ent

34. Synonyms of "Apathy":

A. Passion

B. Concern

C. Empathy

D. Indifference

35. Correct spelling:

A. Beginning

B. Begining

C. Beggining

D. Begening

36. To rain cats and dogs means:

A. Light rain

B. Windy day

C. Heavy rain

D. Foggy condition

37. Which of the following has the same consonant sound as the capitalized part in "rUSh"?

A. Rough

B. Sure

C. Put

D. Plough

38. Correct spelling:

A. Refinary

B. Reffinery



- C. Refinery
D. Refinnery
39. Antonym of "Aberrant":
A. Irregular
B. Abnormal
C. Unusual
D. Normal
40. Correct spelling:
A. Occurence
B. Occurrence
C. Ocurence
D. Occurene
41. Antonym of "Suspect" (again for reinforcement):
A. Trust
B. Accuse
C. Mistrust
D. Doubt
42. Correct spelling:
A. Seargent
B. Sargant
C. Sergeant
D. Sargent
43. Synonyms of "Determination":
A. Weakness
B. Doubt
C. Persistence
D. Laziness
44. To be in the red means:
A. To be rich
B. To be bankrupt
C. To be famous
D. To save money
45. Correct spelling:
A. Colleagues
B. Collegues
C. Coleagues
D. Collagues
46. Correct spelling:
A. Embrassement
B. Embarrassment
C. Embarasment
D. Embarrasement
47. What is the stressed syllable in Democracy?
A. DE-mocracy
B. demo-CRA-cy
C. de-mo-CRACY
D. de-MO-cracy
48. They got ___ the plane at exactly 10:30 p.m.
A. in
B. on
C. into
D. at [2022]
49. *Antonyms*: Batewoi goes about in dowdy clothes, but Chinere puts on ___ clothes.
A. foreign
B. unfashionable
C. fashionable
D. local [2022]
50. *Synonyms*: He got to a path that was narrow.
A. Thin
B. Slim
C. Tiny
D. Lean



Biology Verified Answers and Reasons

1. C

Reason: Chloroplasts are responsible for photosynthesis, converting sunlight into energy, hence referred to as the "kitchen house" of the cell.

2. A

Reason: Lumbar vertebrae are large and strong, supporting most of the body's weight and preventing sagging of the vertebral column.

3. A

Reason: The nucleus controls all cell activities and plays a central role in cell specialization by regulating gene expression.

4. B

Reason: The pulp cavity contains nerves and blood vessels, making it sensitive to temperature changes.

5. A

Reason: Monera (e.g., bacteria) are prokaryotes and lack a true nucleus.

6. D

Reason: Sea urchins belong to the phylum Echinodermata, not Coelenterata.

7. D

Reason: The bicuspid (mitral) valve is located between the

left atrium and left ventricle of the heart.

8. D

Reason: Manganese (Mn) is a trace element required in small amounts by organisms.

9. C

Reason: The pancreas, gall bladder, and spleen are organs closely associated with the stomach.

10. D

Reason: Tropical rainforest biomes support epiphytes — plants that grow on other trees without soil attachment.

11. A

Reason: Epigeal germination involves the epicotyl growing and raising the cotyledons above ground.

12. A

Reason: Hypermetropia (farsightedness) occurs when the eyeball is too short, causing difficulty seeing nearby objects.

13. A

Reason: Converging (convex) lenses help correct presbyopia by aiding near vision.

14. B

Reason: The ability to taste PTC is a discrete trait — one either can or cannot — hence it is



- discontinuous variation.
15. B
Reason: Phenotype refers to the observable traits expressed by a gene.
16. A
Reason: Pathology is the study of diseases and is not considered direct evidence for evolution.
17. A
Reason: Aquatic plants typically have reduced or poorly developed roots since they absorb nutrients directly from water.
18. D
Reason: The cornea and aqueous humor play the primary role in refracting (bending) light entering the eye.
19. A
Reason: The epiglottis closes over the glottis during swallowing to prevent food from entering the trachea.
20. A
Reason: Rhizoids are root-like structures found in bryophytes, not ferns.
21. B
Reason: Karyokinesis is the division of the nucleus during cell division.
22. D
Reason: Both Baobab and Acacia are characteristic trees found in the savannah biome.
23. C
Reason: Bony fish have a swim bladder for buoyancy, which cartilaginous fish lack.
24. D
Reason: Tundra biome has harsh conditions leading to stunted and woody plant growth.
25. A
Reason: Mango is a drupe; tomato is a berry — both are true fruits but of different types.
26. B
Reason: The asexual stage of Plasmodium occurs in humans, particularly in the liver and red blood cells.
27. A
Reason: The stamen is the male part, and the pistil (carpel) is the female part of a flower.
28. D
Reason: Mangrove swamp plants have pneumatophores (breathing roots) to help in gaseous exchange.
29. C
Reason: Anthocyanins are water-soluble pigments responsible for red, purple, and blue



coloration in plants.

30. D

Reason: Perspiration (sweating) helps cool the body by evaporation of sweat from the skin surface.

31. C

Reason: Hepatitis affects the liver, not the kidney.

32. D

Reason: Blood group O- lacks A, B, and Rh antigens, making it the universal donor.

33. D

Reason: Culturing is the process of growing microorganisms under controlled laboratory conditions.

34. D

Reason: Topography (the arrangement of natural and artificial physical features) largely shapes a geographic landscape's profile.

35. B

Reason: Imbibition is the process of water absorption by dry seeds, initiating germination.

36. C

Reason: Hydra belongs to the phylum Coelenterata (also known as Cnidaria).

37. A

Reason: Meiosis reduces the

chromosome number from diploid ($2n$) to haploid (n).

38. C

Reason: The pulp cavity houses nerves and blood vessels in the tooth.

39. C

Reason: Paramecium feeds using the oral groove, which guides food into the cell.

40. C

Reason: Planula is the free-swimming larval stage of cnidarians such as Hydra.

41. B

Reason: Palm trees are monocots and do not undergo secondary growth (increase in thickness).

42. C

Reason: The pericardium is a double-walled membrane that encloses and protects the heart.

43. C

Reason: Kwara State is located in the Guinea savanna zone of Nigeria.

44. B

Reason: *Ascaris lumbricoides* is a parasitic roundworm that can infest the human digestive tract.

45. D

Reason: Co-evolution occurs when two or more species

influence each other's evolution, as seen in flowers and their pollinators.

46. C

Reason: The radula is a toothed, chitinous ribbon used by snails for scraping or cutting food.

47. B

Reason: Anaerobic respiration in plants produces ethanol and carbon dioxide.

48. A

Reason: The nucleus controls gene expression, which determines the cell's specialization.

49. D

Reason: Binocular vision allows the eyes to work together to view objects, giving depth perception and enabling the brain to merge two images.

50. D

51. Answer: C

Reason:

The red coloration of soil is due to the presence of ferric oxide (Fe_2O_3), which gives it a reddish hue.

52. Answer: D

Reason:

Guard cells contain a nucleus, mitochondria, and chloroplasts, but

they lack centrioles, which are typically absent in plant cells.

53. Answer: D

Reason:

Langerhans cells are specialized immune cells in the skin that present antigens and initiate immune responses.

54. Answer: C

Reason:

Cassava mosaic disease is caused by a virus, specifically geminiviruses.

55. Answer: C

Reason:

Mucor and Rhizopus absorb carbohydrate in the form of glucose, a simple sugar.

56. Answer: B

Reason:

Palm trees are monocots and do not undergo secondary growth, unlike dicot trees like mango and mahogany.

57. Answer: A

Reason:

The capsule is not part of a fern's prothallus. It is found in mosses. The prothallus contains archegonia, antheridia (not androecium), and



rhizoids.

58. Answer: C

Reason:

Abscission refers to the natural shedding of leaves, fruits, or flowers from a plant.

59. Answer: B

Reason:

Agama agama (common agama lizard) exhibits territorial behavior.

60. Answer: B

Reason:

The axon carries impulses away from the nerve cell body.

61. Answer: D

Reason:

Bemisia nigeriensis, a whitefly species, transmits cassava mosaic virus.

62. Answer: C

Reason:

Onions store food in their leaves, which are thick and fleshy for storage.

63. Answer: C

Reason:

Mucor or Rhizopus have chitin, not cellulose, in their cell walls, so option C is incorrect.

64. Answer: A

Reason:

The advancement of ferns over mosses is supported by comparative anatomy, showing vascular tissues in ferns.

65. Answer: B

Reason:

Abscissic acid (ABA) regulates the opening and closing of stomata, particularly in response to stress.

66. Answer: C

Reason:

The exocarp is part of a fruit, not the seed. The seed includes the testa (seed coat), endosperm, and embryo.

67. Answer: B

Reason:

The tse-tse fly transmits Trypanosoma brucei, the parasite responsible for trypanosomiasis (sleeping sickness).

Chemistry verified Answers and Reasons

1. A

Reason: Carbon dioxide (CO₂) is a greenhouse gas responsible for global warming due to its heat-trapping capability in the



atmosphere.

2. B

Reason: The trivial name for butadenoic acid is maleic acid.

3. C

Reason: Molanic acid is another name for ethanedioic acid (oxalic acid).

4. D

Reason: Petroleum coke is residual carbon obtained during the cracking of petroleum.

5. D

Reason: Coal fibre is not produced during destructive distillation of coal. The main products are coke, coal tar, ammoniacal liquor, and coal gas.

6. A

Reason: Hydrogen chloride (HCl) is highly soluble in water and cannot be collected over it.

7. C

Reason: Tool steel contains iron, carbon, and tungsten to improve hardness and durability.

8. C

Reason: Vanadium(V) oxide (V_2O_5) is the catalyst used in the Contact Process for making sulfuric acid.

9. C

Reason: Propane is a saturated hydrocarbon and does not

contain a double bond.

10.C

Reason: The Tyndall effect is the scattering of light by particles in a colloid.

11.B

Reason: The C=O bond in carbonyl compounds is covalent in nature.

12.A

Reason: The correct order of increasing oxidation strength is $F > Cl > Br > I$.

13.A

Reason: The formula for water glass is sodium metasilicate (Na_2SiO_3).

14.D

Reason: When calcium compounds are heated in a flame, they emit a bright light called lime light.

15.C

Reason: In the extraction of aluminum from bauxite, NaOH is used to dissolve alumina.

16.B

Reason: The IUPAC name for capric acid is decanoic acid.

17.A

Reason: Vapor density = molar mass / 2, hence molar mass = $15 \times 2 = 30$ g/mol. The alkane with this mass is ethane (C_2H_6).



18.B

Reason: Carbon fibre is used in making plastic composites due to its strength and lightness.

19.A

Reason: Finely divided iron is the catalyst used in the Haber process for ammonia synthesis.

20.C

Reason: V_2O_5 is used as a catalyst in the conversion of SO_2 to SO_3 in the Contact Process.

21.D

Reason: Sodium bicarbonate ($NaHCO_3$) decomposes on heating to give CO_2 and Na_2CO_3 .

22.C

Reason: Gamma rays are used for sterilization of hospital equipment due to their high energy.

23.A

Reason: Oxygen is only slightly soluble in water and can be collected over it.

24.B

Reason: Formic acid (methanoic acid) occurs naturally in red ants.

25.B

Reason: Lattice energy increases with decreasing ionic size; correct order is $NaI < NaBr < NaCl < NaF$.

26.A

Reason: Salt cake soda is sodium sulfate (Na_2SO_4).

27.D

Reason: Halogen reactivity decreases down the group, not increases.

28.A

Reason: Molar mass of $CH_6O = 32 \text{ g/mol}$; $64 / 32 = 2$; molecular formula = $(CH_6O)_2 = C_{12}H_{12}O_2$.

29.C

Reason: Photographic flashlights use krypton and xenon gases for intense flashes.

30.C

Reason: Ideal gas behavior is approximated at high temperature and low pressure.

31.A

Reason: $K_{sp} = [Ag^+][Cl^-] = (1.35 \times 10^{-5})^2 = 1.82 \times 10^{-10} \text{ mol}^2/\text{dm}^6$.

32.D

Reason: Radon is the only radioactive noble gas naturally found.

33.B

Reason: For MX_2 : $K_{sp} = 4s^3 = 6.0 \times 10^{-9} \rightarrow s = \text{cube root of } (6.0/4 \times 10^{-9}) \approx 3 \times 10^{-5} \text{ mol/dm}^3$.

34.A

Reason: Sea divers use helium and oxygen to avoid nitrogen narcosis.



35.C

Reason: Xenon is used in aerodrome beacons due to its bright white light emission.

36.B

Reason: The normal pH of human blood is 7.4.

37.C

Reason: Argon is the most abundant noble gas in Earth's atmosphere.

38.C

Reason: Phosphorus makes up about 0.32% of the human body by mass.

39.A

Reason: Iron (Fe) forms only basic oxides, not amphoteric ones.

40.A

Reason: Radon is used in the treatment of certain cancers due to its radioactivity.

41. Answer: A

Reason:

The confirmatory test for water is that anhydrous cobalt(II) chloride paper turns from blue to pink or white in the presence of water.

42. Answer: A

Reason:

Sodium hydroxide is used to test for metal ions such as Ca^{2+} , Zn^{2+} , Cu^{2+} ,

and NH_4^+ . All listed ions are tested using NaOH, but Ca^{2+} forms a white precipitate that is only slightly soluble in excess, making it a key identification.

43. Answer: D

Reason:

Coal fiber is not a product of the destructive distillation of coal. Products like coal tar, benzene, and ammoniacal liquor are.

44. Answer: C

Reason:

The general molecular formula of alkanes is $\text{C}_n\text{H}_{2n+2}$.

45. Answer: C

Reason:

Ammonia gas molecules exhibit both translational (movement from place to place) and vibrational motion.

46. Answer: B

Reason:

Formalin is an aqueous solution of 37% formaldehyde and 63% water.

47. Answer: D

Reason:

Ethanoic acid (acetic acid) is used in the production of some blood



pressure medications due to its physiological effects.

48. Answer: C

Reason:

Saturated vapour pressure (SVP) primarily depends on temperature.

49. Answer: C

Reason:

Sodium argentocyanide solution is the electrolyte used in silver electroplating.

50. Answer: C

Reason:

Aluminium is used in thermite welding because it is a strong reducing agent that reduces metal oxides to the metal.

51. Answer: D

Reason:

Solder is an alloy of tin and lead or tin and zinc in some cases.

52. Answer: D

Reason:

Stainless steel is made primarily from iron, chromium, carbon, and nickel. It does not contain aluminium.

53. Answer: D

Reason:

Duralumin is an alloy of aluminium, not copper. Brass and bronze are copper-based alloys.

54. Answer: D

Reason:

The Haber Process is used for the industrial production of ammonia (NH_3).

55. Answer: B

Reason:

Methanol is also known as wood spirit.

56. Answer: B

Reason:

Bronze is an alloy of copper and tin.

57. Answer: D

Reason:

C_3H_8 (propane) is a saturated hydrocarbon with only single bonds.

58. Answer: C

Reason:

Groundnut oil and water can be separated by difference in density (oil floats on water).

59. Answer: B

Reason:

In electrolytic refining, the impure



metal is made the anode.

60. Answer: D

Reason:

Malonic acid is also known as propanedioic acid.

Physics Verified Answers and Reasons

1. B

Reason: Floatation involves a balance between the upthrust (buoyant force) and the weight of the object.

Important Note: If upthrust > weight, the object rises; if equal, it floats.

2. C

Reason: The anomalous expansion of water (expands below 4°C) was discovered by Hope.

Mnemonic: *Hope* discovered the *Hope's apparatus* related to water's anomaly.

3. B

Reason: Damping refers to the gradual loss of energy of an oscillating system until it comes to rest.

Important Note: Damping reduces amplitude over time.

4. C

Reason: A spring balance operates based on Hooke's law: extension is proportional to force.

Important Note: $F = kx$, where k is spring constant.

5. D

Reason: A battery inverter steps up a

low DC voltage (from the battery) to a higher AC voltage.

Important Note: It's commonly used in fuel-powered generators for appliances.

6. C

Reason: Acceleration = derivative of velocity.

Given $V = 5 + 3t^2$, $dV/dt = 6t \rightarrow$ at $t = 2s$,
 $a = 6 \times 2 = 12 \text{ m/s}^2$.

7. B

Reason: Weight $\propto$ density for a fixed volume.

New weight = $(800/900) \times 60 \text{ g} = 53.3 \text{ g}$.

8. A

Reason: Power = Work/Time = (Force $\times$ Distance)/Time = $(MLT^{-2} \times L)/T = ML^2T^{-3}$.

Mnemonic: Power = ML^2T^{-3}

9. A

Reason: A transformer operates on the principle of electromagnetic induction (Faraday's law).

Important Note: It requires a varying magnetic field (AC current).

10. B

Reason: Moving iron meters can measure both AC and DC.

Important Note: Galvanometer is typically DC only.

11. A

Reason: Air is a poor conductor of heat, hence acts as an insulator.

Important Note: Air gaps are used in



thermos flasks for insulation.

12. C

Reason: Micrometer screw gauge is best for measuring very small thicknesses like paper.

Important Note: Accuracy up to 0.01 mm.

13. C

Reason: Doping silicon with phosphorus adds free electrons → N-type semiconductor.

Mnemonic: Phosphorus = 5 valence electrons → donates electrons → N-type.

14. Answer: D

Reason:

The resultant of vectors is maximum when the vectors are in the same direction, i.e., the angle between them is 0° .

15. Answer: B

Reason:

The resultant of two vectors is minimum when they are in opposite directions, i.e., anti-parallel (angle between them is 180°).

16. Answer: C

Reason:

An electric bell works using electromagnets, solenoids, and a contact breaker. It does not use a

permanent magnet.

17. Answer: C

Reason:

Work done = Charge $\times$ Potential difference = $10\text{ C} \times 12\text{ V} = 120\text{ J}$

18. Answer: B

Reason:

Relative velocity is the velocity of one object with respect to another object.

19. Answer: C

Reason:

Mechanical Advantage = Effort arm / Load arm = $2.0\text{ m} / 0.5\text{ m} = 4.0$

20. Answer: B

Reason:

From $80 \rightarrow 40 \rightarrow 20 \rightarrow 10$ (3 halvings).

Time = 3 hours $\Rightarrow$ Half-life = 1 hour

21. Answer: B

Reason:

Specific heat capacity has the unit: $\text{J/kg}^\circ\text{C}$

22. Answer: B

Reason:

Gamma rays have the highest frequency among electromagnetic waves.

23. Answer: B

Reason:

Pressure = $pgh = 1000 \times 10 \times 10 =$

100,000 Pa

24. Answer: A

Reason:

$$Q = mc\Delta T = 2 \times 400 \times (80 - 20) = 48,000 \text{ J}$$

25. Answer: A

Reason:

$64 \rightarrow 32 \rightarrow 16 \rightarrow 8$ (3 half-lives = 15 days). Final mass = 8 g

26. Answer: D

Reason:

Volume does not directly affect the rate of heat transfer.

27. Answer: C

Reason:

Pressure in a fluid at rest acts equally in all directions at the same depth.

28. Answer: A

Reason:

Standard value for gravity used in calculations is 10 m/s^2

29. Answer: A

Reason:

$$\text{Density} = \text{mass/volume} = 500 \text{ g} / 200 \text{ cm}^3 = 2.5 \text{ g/cm}^3$$

30. Answer: D

Reason:

$$MA = \text{Efficiency} \times VR = 0.8 \times 5 = 4.0$$

31. Answer: A

Reason:

$$\lambda = v/f = 3 \times 10^8 / 1.5 \times 10^6 = 200 \text{ m}$$

32. Answer: C

Reason:

Interference is a characteristic of wave behavior and supports wave nature.

33. Answer: C

Reason:

$$\text{Energy} = \text{Power} \times \text{Time} = 60 \text{ W} \times 5 \text{ h} = 300 \text{ Wh}$$

34. Answer: A

Reason:

Using heat balance, final temp $\approx 54^\circ\text{C}$. Closest option is 50°C .

35. Answer: C

Reason:

Gravitational field strength decreases with distance from Earth's surface.

36. Answer: B

Reason:

AC current alternates in direction and magnitude.

37. Answer: B

Reason:

$$\text{Pressure} = \text{Force} / \text{Area} = 100 / 2 = 50 \text{ N/m}^2$$

38. Answer: A

Reason:



Resultant downstream velocity = $4 + 3$
= 7 m/s

39. Answer: C

Reason:

The photoelectric effect supports the particle-like nature of light.

40. Answer: A

Reason:

Efficiency = $(200 \times 5) / (100 \times 15) =$
 $1000 / 1500 = 66.7\%$

41. Answer: B

Reason:

Inductive reactance has the unit Ohms (Ω)

42. Answer: A

Reason:

$f = E/h = 6.6 \times 10^{-19} / 6.6 \times 10^{-34} = 1 \times 10^{15}$
Hz

43. Answer: B

Reason:

Total resistance in series = $2 + 3 + 5 =$
 10Ω

44. Answer: B

Reason:

Latent heat = $Q/m = 336,000 / 1 = 3.36$
 $\times 10^5 \text{ J/kg}$

45. Answer: C

Reason:

$P = I^2 R = 4^2 \times 5 = 80 \text{ W}$

46. Answer: A

Reason:

Pressure is a scalar — it has magnitude but no direction.

47. Answer: A

Reason:

$V = IR = 2 \times 3 = 6 \text{ V}$

48. Answer: D

Reason:

An ideal machine has 100% efficiency — no energy is lost.

49. Answer: D

Reason:

$1/16 = 4 \text{ half-lives} \Rightarrow 4 \times 2 = 8 \text{ years}$

50. Answer: B

Reason:

Electromagnetic waves are transverse in nature.

English Verified Answers and Reason

1. B. Guerrilla

Reason: Correct spelling.

'Guerrilla' refers to a member of a small independent group taking part in irregular fighting.

2. A. Maintenance

Reason: Correct spelling.

3. C. To watch carefully

Reason: "To look out for" means



- to be vigilant or watchful for something or someone.
4. B. Unstoppable force
Reason: "Juggernaut" refers to a huge, powerful, and overwhelming force.
5. A. Disputable
Reason: Antonym of "Undeniable" is something that can be denied or questioned.
6. A. Hippopotamus
Reason: Correct spelling.
7. C. ad-VER-tise-ment
Reason: In British English, the stress is on the second syllable.
8. A. Correspondence
Reason: Correct spelling.
9. C. Disrupted
Reason: "Derailed" means thrown off course or disrupted.
10. A. Weakness
Reason: Antonym of "Determination" is a lack of resolve or strength.
11. B. PRAC-ti-cal
Reason: Stress is on the first syllable.
12. B. Applicable
Reason: Synonym of "Pertinent" is relevant or applicable.
13. D. Aggression
Reason: Correct spelling.
14. B. demo-CRA-tic
Reason: Stress is on the third syllable.
15. C. Scolded
Reason: "Reprimanded" means to be formally scolded.
16. A. To be boring
Reason: "Tedious" means monotonous or dull.
17. A. Depress
Reason: Antonym of "Stimulate" is to lower or discourage activity.
18. C. accom-mo-da-TION
Reason: Stress is on the fourth syllable before "-tion".
19. B. Surprise
Reason: Correct spelling.
20. D. She
Reason: The "sh" sound in "bash" is the same in "she".
21. A. Mischievous
Reason: Correct spelling. Note: Pronounced as /MIS-chuh-vuhs/, not /MIS-chee-vee-us/.
22. A. Brassiere
Reason: Correct spelling for a woman's undergarment.
23. B. Absolve
Reason: Antonym of "Condemn" is to forgive or clear of blame.
24. B. To be angry
Reason: "To see red" is an idiom meaning to be furious.
25. D. om-ni-VOR-ous
Reason: Stress is on the third syllable.

26. A. Corroborate

Reason: Correct spelling.

27. A. Thoughtful

Reason: Antonym of

"Impetuous" is someone who acts with consideration.

28. A. Insistence

Reason: "Emphasis" means special importance or insistence.

29. A. Unstable

Reason: "Erratic" means not regular or unpredictable.

30. D. ju-DI-ciary

Reason: Stress is on the second syllable.

31. A. Trust

Reason: Antonym of "Suspect" is to believe or trust.

32. A. Authentic

Reason: Correct spelling.

33. A. om-NI-scient

Reason: Stress is on the second syllable.

34. D. Indifference

Reason: Synonym of "Apathy" is lack of interest or concern.

35. A. Beginning

Reason: Correct spelling.

36. C. Heavy rain

Reason: "To rain cats and dogs" means to rain very heavily.

37. B. Sure

Reason: "sh" sound in "rush" matches that in "sure".

38. C. Refinery

Reason: Correct spelling.

39. D. Normal

Reason: Antonym of "Aberrant" is standard or typical.

40. B. Occurrence

Reason: Correct spelling.

41. A. Trust

Reason: Reinforcement: Antonym of "Suspect" is "Trust".

42. C. Sergeant

Reason: Correct spelling.

43. C. Persistence

Reason: Synonym of "Determination" is resolve or perseverance.

44. B. To be bankrupt

Reason: "In the red" means owing money or in debt.

45. A. Colleagues

Reason: Correct spelling.

46. B. Embarrassment

Reason: Correct spelling.

47. B. demo-CRA-cy

Reason: Stress is on the third syllable.

48. B. on

Reason: We say "get on the plane," not "in."

49. C. fashionable

Reason: Opposite of "dowdy" is fashionable.

50. A. Thin

Reason: "Narrow" is best

matched with "thin" as a
synonym.

